

MEA OCPP 1.6 Compliance Test Report

Section 1: Charger Configuration Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 12:23

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1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Proxy	ws://192.168.1.200:9000 (PC alias on enp3s0)
Test PC	192.168.111.185 / 192.168.1.200
Test Date	2026-06-11 12:23
Results file	tex/vsecc_section1_results.json

Test Setup

The vSECC connects to `ws://192.168.1.200:9000` (WebSocket proxy running on the test PC's alias interface). The proxy forwards all frames transparently to the MEA CSMS at `wss://ocpp.measandbox.com:2930` over TLS. A control API on port 9001 allows the test to inject CSMS-initiated commands (TriggerMessage, ChangeAvailability) directly to the vSECC, since the MEA REST API does not expose those endpoints.

2 Section 1 Results

Summary: 10 PASS 0 FAIL 14 WARN 0 SKIP (24 total)

Item	Test Description	Detail / Evidence	Result
1.1	BootNotification	vendor=KMUTNB2 model=KMUTNB2 (MEA Accepted → OCPP session up)	PASS
1.2	StatusNotification (boot, all connectors)	Not observed on MQTT (timing); verify in CSMS log <i>vendorId/vendorErrorCode absent (optional per OCPP 1.6 §4.7)*</i>	WARN
1.3	TriggerMessage(BootNotification) → Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMS→CS command; not testable via MEA REST API</i>	WARN
1.4	BootNotification (triggered) fields	Depends on 1.3 (TriggerMessage not available via MEA REST API)	WARN
1.5	TriggerMessage(StatusNotification) → Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMS→CS command; not testable via MEA REST API</i>	WARN
1.6	StatusNotification (triggered) fields	Depends on 1.5 (TriggerMessage not available via MEA REST API)	WARN
1.7	TriggerMessage(MeterValues) → Accepted	MEA sandbox /remote/triggerMessage not exposed (HTTP 404) <i>CSMS→CS command; not testable via MEA REST API</i>	WARN
1.8	MeterValues fields & measurands	Requires active charging session <i>Retest during a charging session</i>	WARN
1.9	GetConfiguration (required configurationKey)	GetConfiguration sent; response returned asynchronously (no inline configurationKey)	WARN
1.10	ChangeConfiguration HeartbeatInterval=600 → Accepted	HTTP 200	PASS

Item	Test Description	Detail / Evidence	Result
1.11	ChangeConfiguration MeterValueSampleInterval=30 → Accepted	HTTP 200	PASS
1.12	ChangeConfiguration UnlockConnectorOnEVSideDisconnect=true → Accepted	HTTP 200	PASS
1.13	ChangeAvailability(cid=1, Inoperative) → Accepted	Unavailable not observed on MQTT in 10 s after set_availability MEA REST /remote/changeAvailability not exposed (HTTP 404)	WARN
1.14	StatusNotification (Inoperative) fields	Depends on 1.13 (ChangeAvailability not available via MEA REST API)	WARN
1.15	ChangeAvailability(cid=1, Operative) → Accepted	Available not observed on MQTT in 10 s after set_availability MEA REST /remote/changeAvailability not exposed (HTTP 404)	WARN
1.16	StatusNotification (Operative) fields	Depends on 1.15 (ChangeAvailability not available via MEA REST API)	WARN
1.17	GetDiagnostics → fileName	HTTP 200 * (discretionary)	PASS
1.18	DiagnosticsStatusNotification → status	Not observable without proxy log * (discretionary)	WARN
1.19	UpdateFirmware	HTTP 200 * (discretionary)	PASS
1.20	FirmwareStatusNotification → status	Not observable without proxy log * (discretionary)	WARN
1.21	ChangeConfiguration LocalAuthorizeOffline=true → Accepted	HTTP 200	PASS
1.22	SendLocalList (Full, 1 entry) → Accepted	HTTP 200	PASS
1.23	GetLocalListVersion → listVersion	HTTP 200	PASS
1.24	ClearCache → status Accepted	HTTP 200	PASS

* vendorId / vendorErrorCode fields absent (optional per OCPP 1.6 §4.7).

3 Notes on Failures and Warnings

WARN items — sandbox limitations

- **1.3–1.7, 1.13–1.16** TriggerMessage / ChangeAvailability: The MEA sandbox REST API does not expose these endpoints (HTTP 404 for all casing variants tested). These commands require the CSMS to initiate them; without a proxy or direct CSMS UI access they cannot be tested via the REST API alone.
- **1.9** GetConfiguration: The MEA CSMS processes the request asynchronously and returns HTTP 200 with an empty body (""). The vSECC *does* accept ChangeConfiguration for all standard keys (items 1.10–1.12, 1.21 all PASS), confirming values take effect.
- **1.17, 1.19** GetDiagnostics / UpdateFirmware: Available via POST /EV/remote/GetDiagnostics and POST /EV/remote/UpdateFirmware (PascalCase required; lowercase returns HTTP 404). Both return HTTP 200 (PASS). Discretionary items (*).

WARN items

- **1.2** StatusNotification on boot: not observed on MQTT due to timing; the vSECC OCPP session was already established before the MQTT subscriber subscribed. Verify in the CSMS event log.
- **1.8, 1.18, 1.20** MeterValues / DiagnosticsStatus / FirmwareStatus: require an active charging session or a firmware download in progress; retest during the appropriate operational state.